

Domain and range



- 1
- a Any value of x .
 - b Only non-negative values of y .
- 2
- a 0
 - b There is no greatest value.
 - c $y \geq 0$
 - d All real x
- 3
- a 0
 - b There is no greatest value.
 - c $y \geq 0$
 - d All real x
- 4 Example answer: $f(x) = |x| + 2$
- 5
- a $[0, 6]$
 - b $(3, 6]$
- 6
- a
 - i All real x
 - ii $y \leq 0$
 - b
 - i All real x
 - ii $y \geq -1$
 - c
 - i All real x
 - ii $y \leq 4$
 - d
 - i All real x
 - ii $y \leq 2$
- 7
- a
 - i $[-5, -2]$
 - ii $[0, 6]$
 - b
 - i $[-2, 3)$
 - ii $[-9, 0]$
 - c
 - i $(-2, 1)$
 - ii $[0, 6)$
 - d
 - i $[2, 4)$
 - ii $(0, 4]$



Graphs of absolute value functions

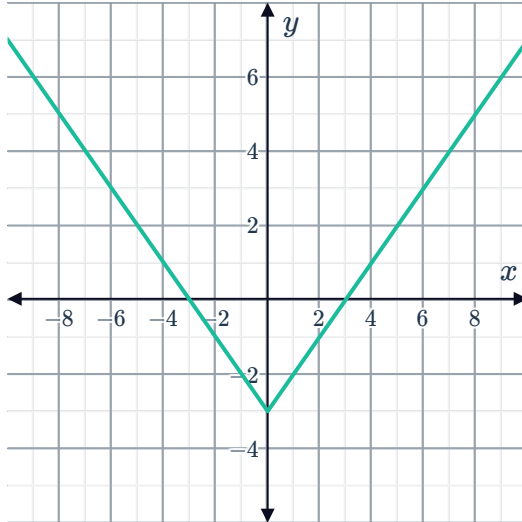
8

a

i All real x

ii $y \geq -3$

iii

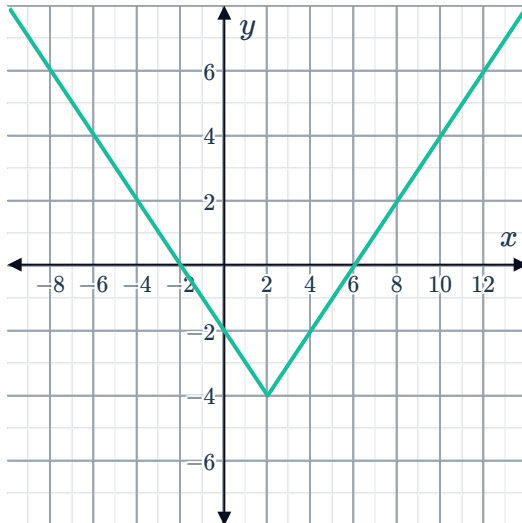


b

i All real x

ii $y \geq -4$

iii



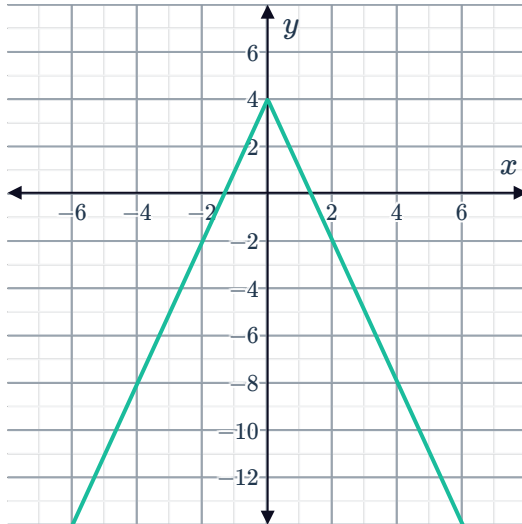
c

i All real x



ii $y \leq 4$

iii

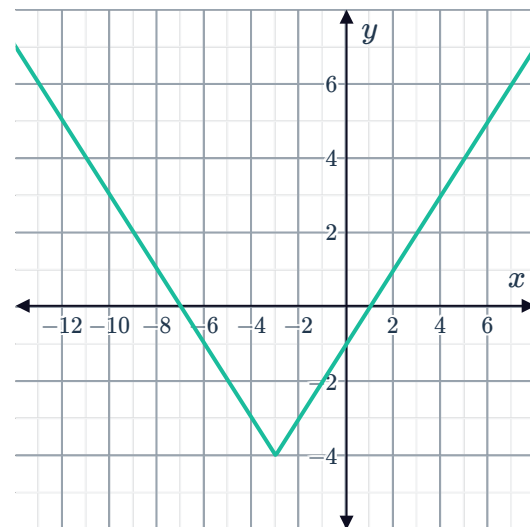


9

a

i $y \geq -4$

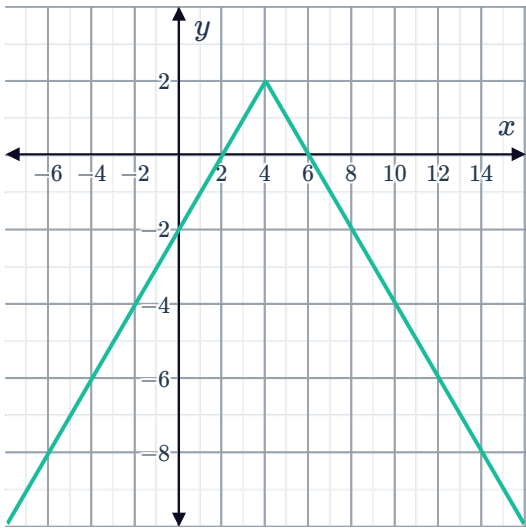
ii



iii $(-3, -4)$

iv -4

b
i $y \leq 2$



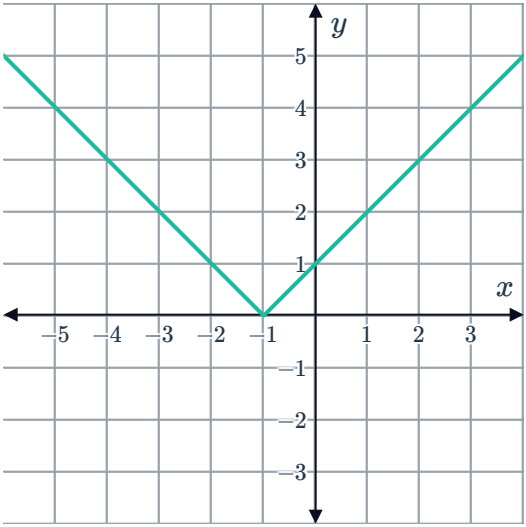
iii (4,2)

iv 2

10
a

x	-2	-1	0	1	2
y	1	0	1	2	3

b



c $x = -1$

d (-1,0)

e

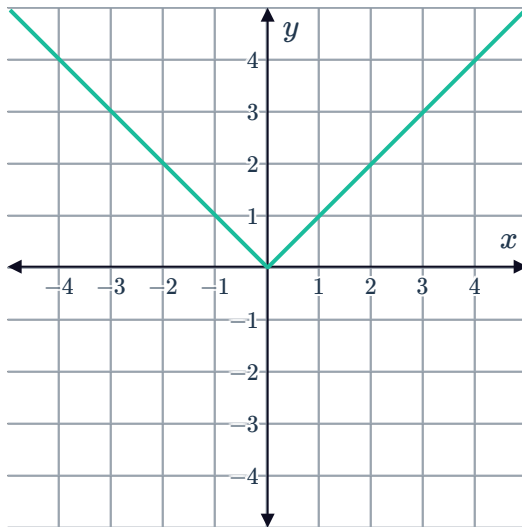
	Equation	Slope
$x < -1$	$y = -x - 1$	-1
$x > -1$	$y = x + 1$	1

11



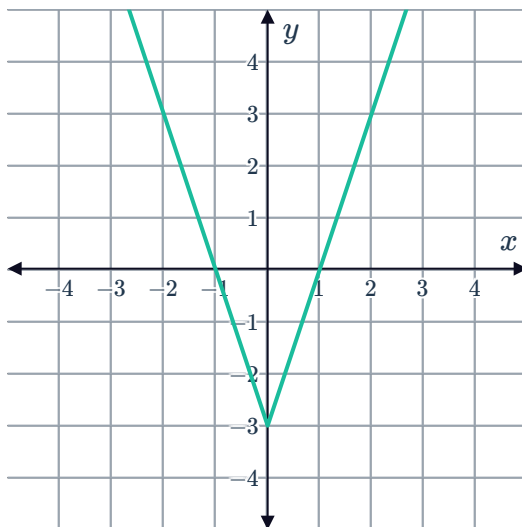
a

i

ii $(0,0)$ iii $-\infty < x < \infty$ iv $y \geq 0$

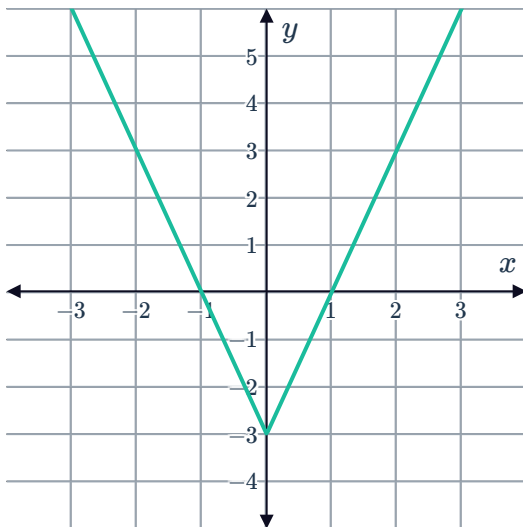
b

i

ii $(0,-3)$ iii $-\infty < x < \infty$ iv $y \geq -3$

12

a



b

$x = 0$

c $(0, -3)$

e $y = -3$

g 3

d $x = -1, 1$

f -3

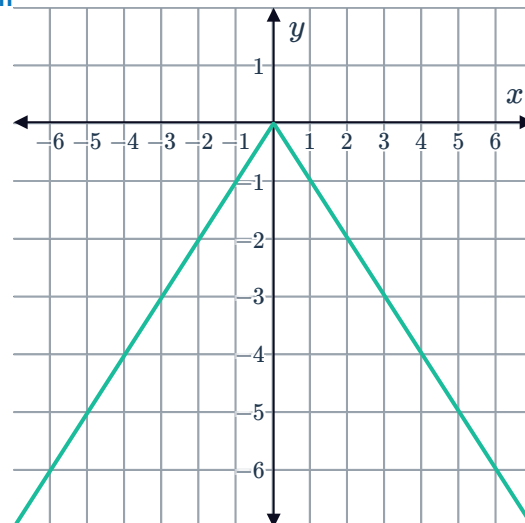
13

a

i Downwards

ii $(0, 0)$

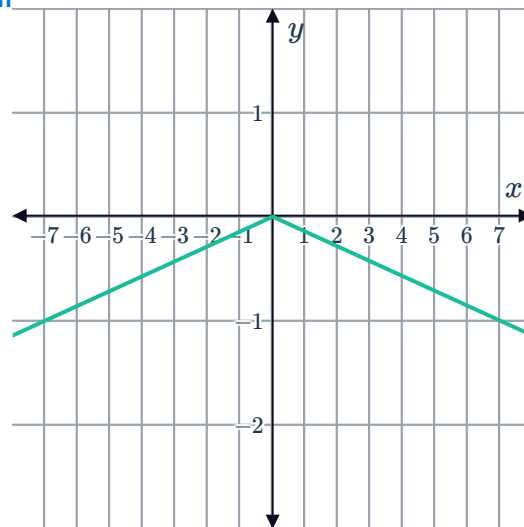
iii



b

i Downwards

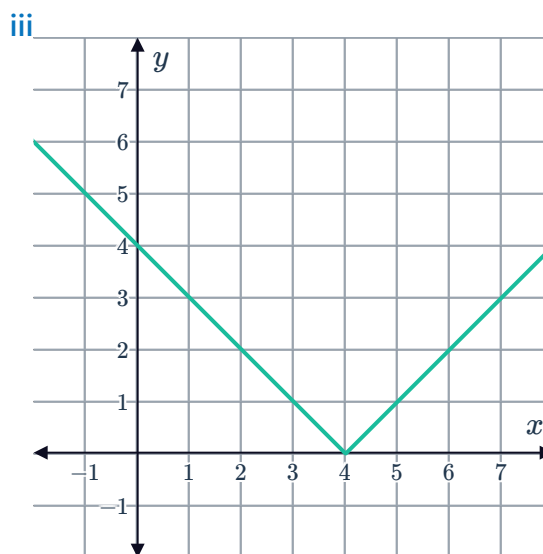
ii $(0,0)$



c

i Upwards

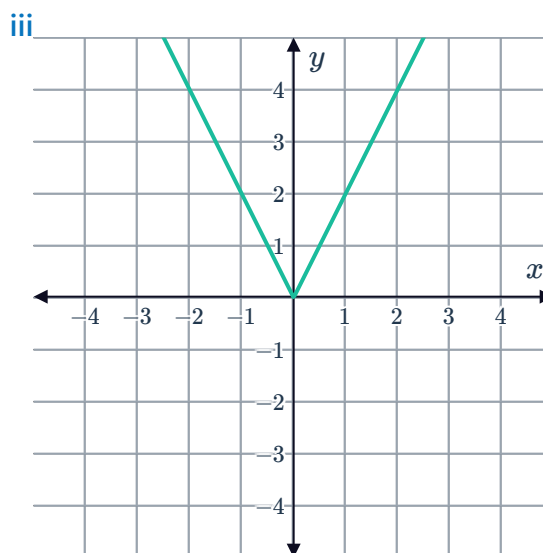
ii $(4,0)$



d

i Upwards

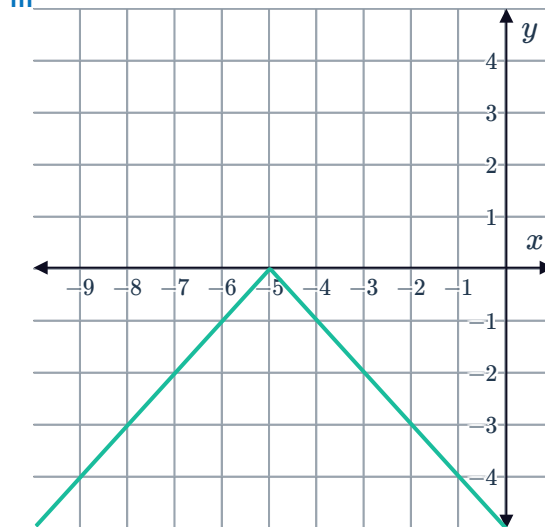
ii $(0,0)$



e

i Downwards

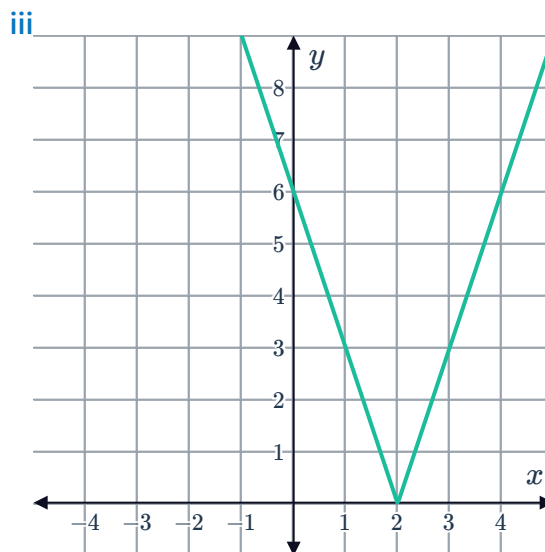
ii $(-5, 0)$



f

i Upwards

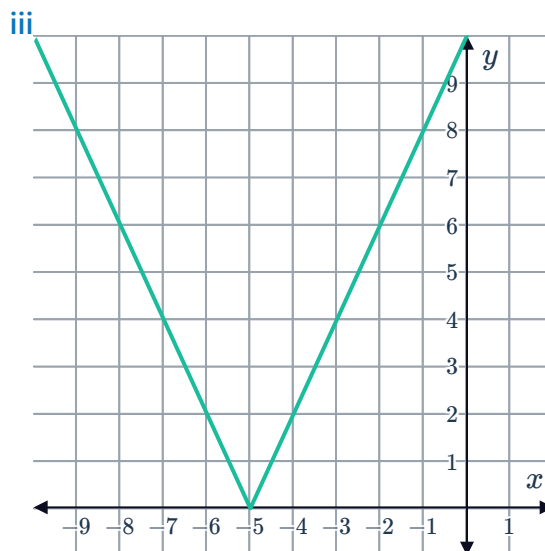
ii $(2, 0)$



g

i Upwards

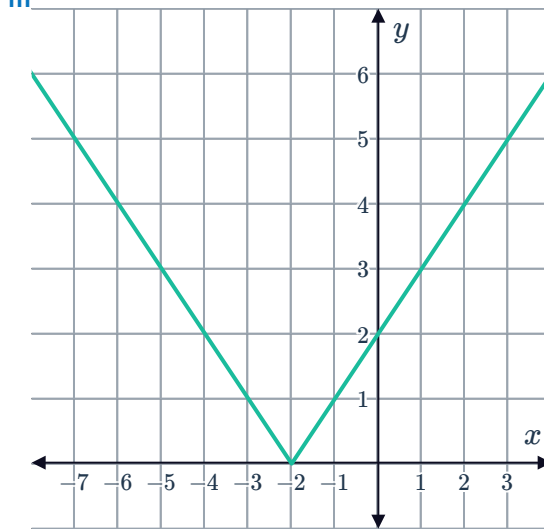
ii $(-5, 0)$



h

i Upwards

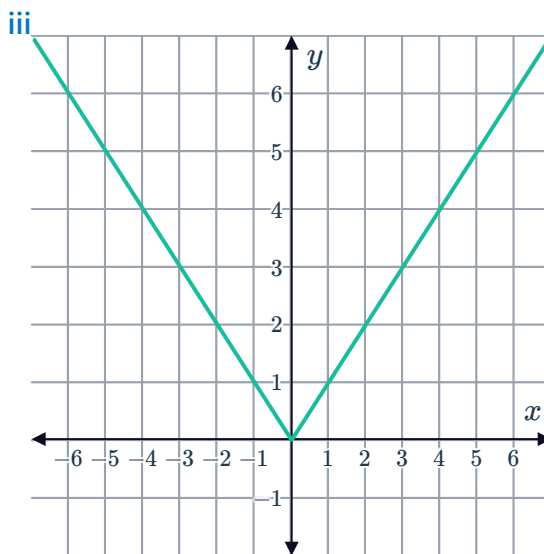
ii $(-2, 0)$



i

i Upwards

ii $(0, 0)$



14

a Upwards

b $(-1, 0)$

c $x = -1$

d

i No

ii No

iii Yes

15

a Upwards

b $(3, 0)$

c $x = 3$

d

i No

ii No

iii Yes

iv No

16

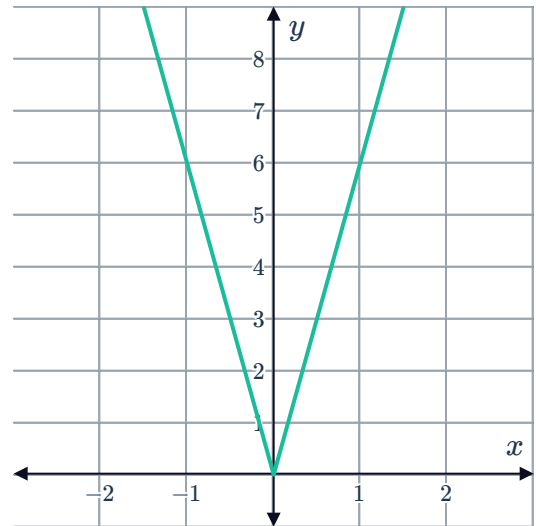
a Upwards

c $(0,0)$



b $x = 0$

d



e

i No ii No iii Yes iv Yes

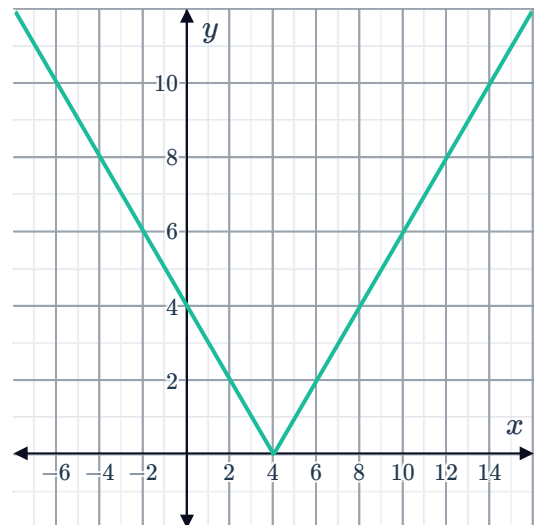
17

a Upwards

c $(4,0)$

b $x = 4$

d



e

i Yes ii No iii No iv Yes

18

a 3

c $(-\infty, \infty)$

e

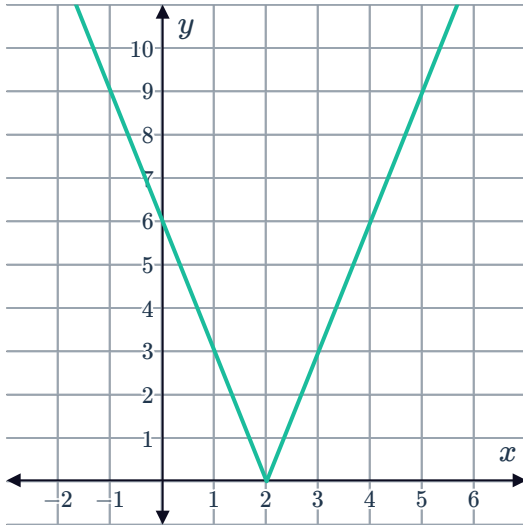
b $f(x) = |3(x - 4)|$

d $[0, \infty)$



19

a

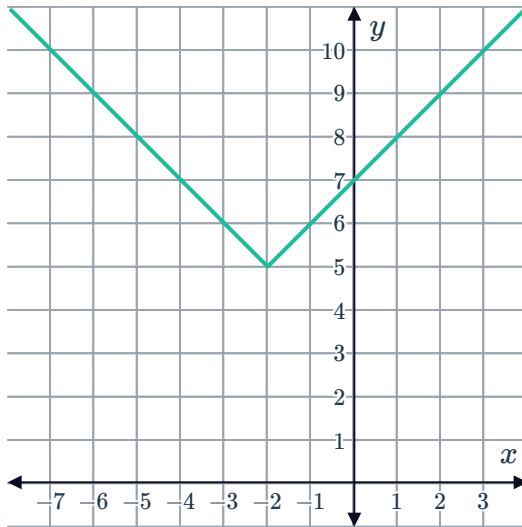


b (2, 0)

20

a

i

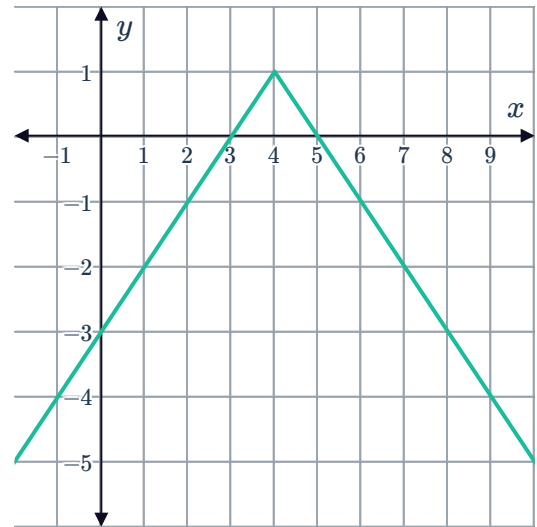


ii (-2, 5)

iii 5

b

i

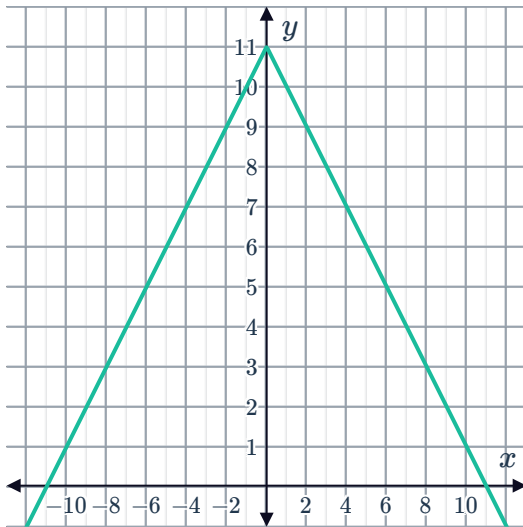


ii (4, 1)

iii 1

21

a

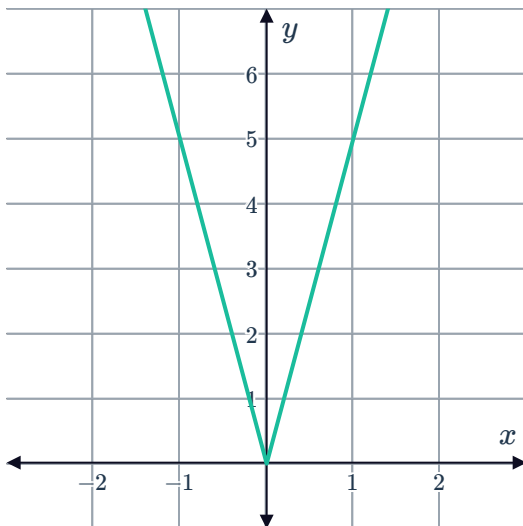


11

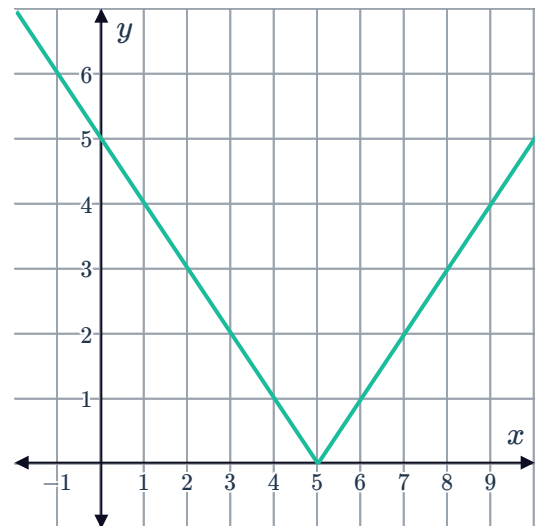
c $(-\infty, 0)$ d $(0, \infty)$

22

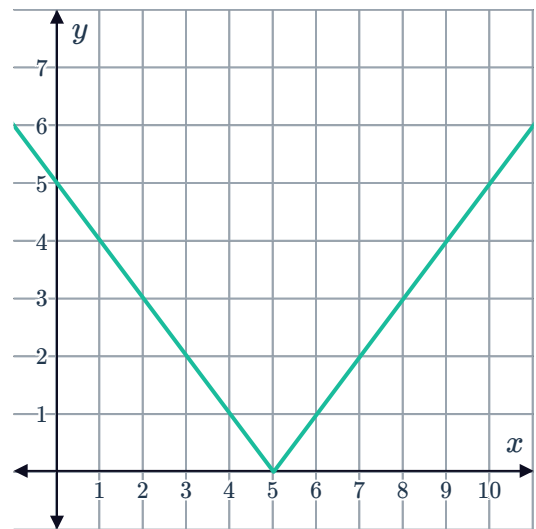
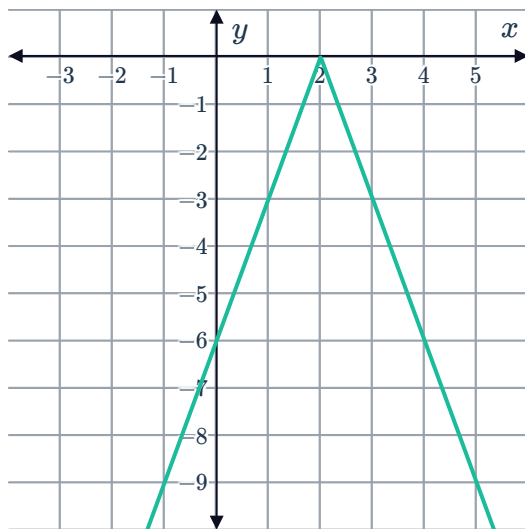
a



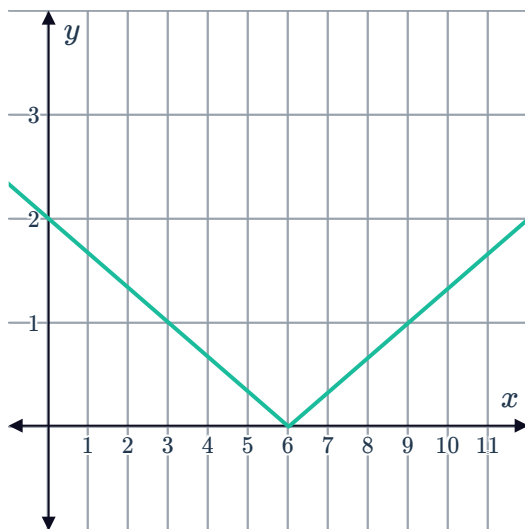
b



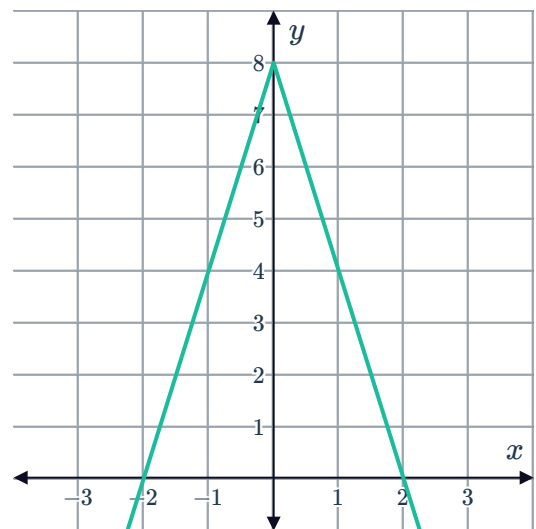
c



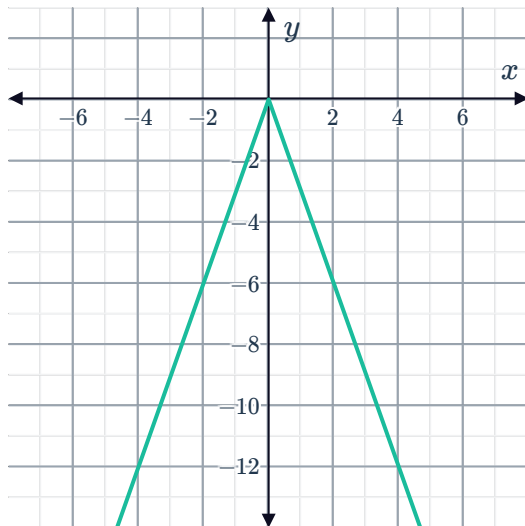
e



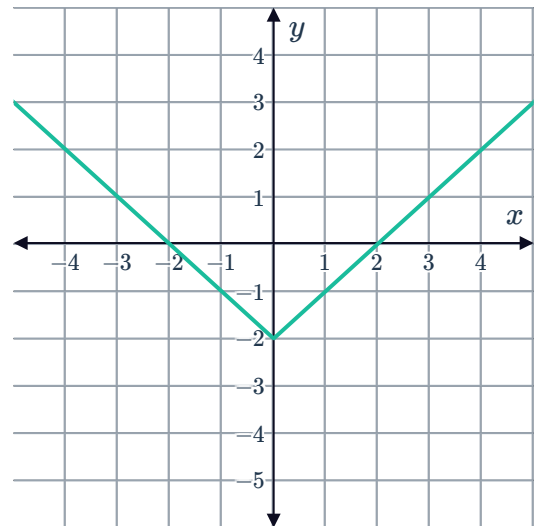
f



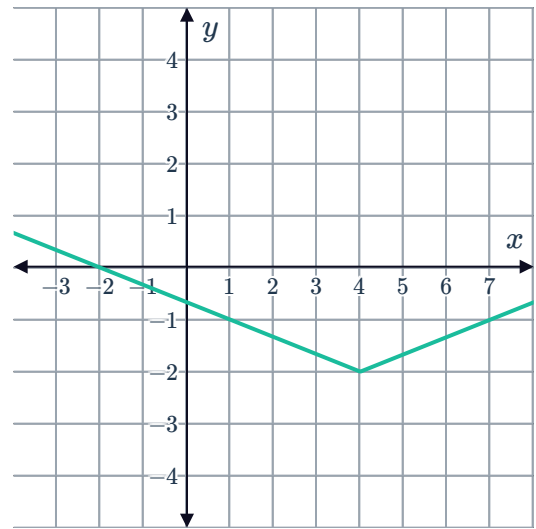
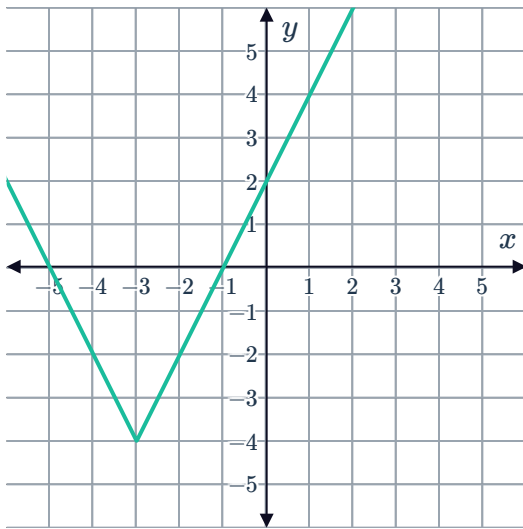
g



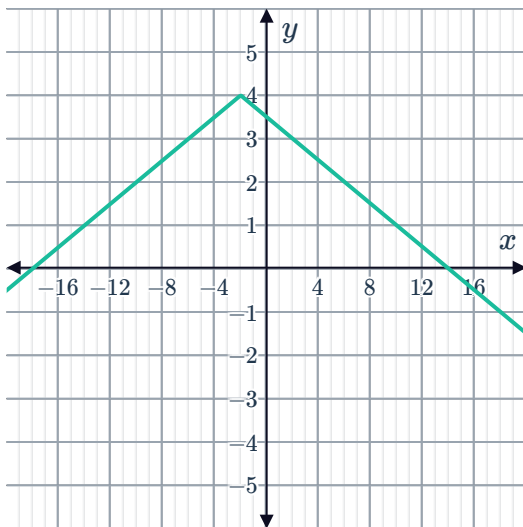
h



i



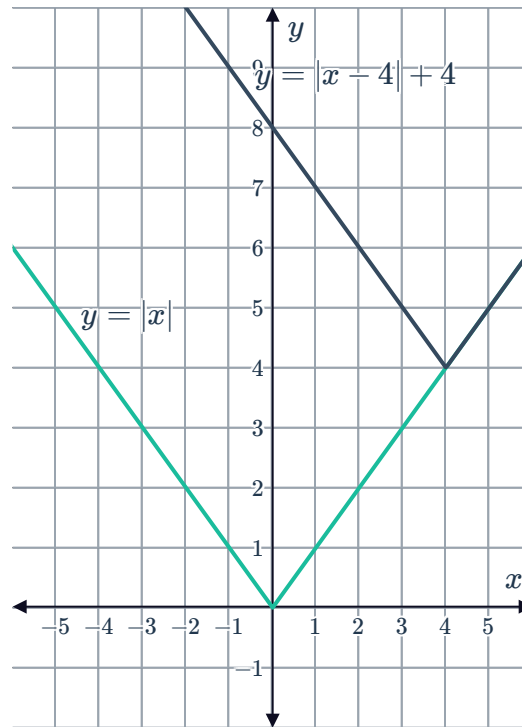
k





Transformations

23



24 $f(x) = |x + 2|$

25

a $y = |x + 5| - 3$

b $y = -4|x|$

c $y = -\frac{|x + 3|}{4} + 3$

26

a Horizontal translation

b Yes

c Yes

27 Translation of 4 units down

28 Vertical stretching by a factor of 3.

29

a Reflection about the x -axis and translation of 4 units up.

b $f(x) = 4 - |x|$

30 a 3



b -3

c $(0, 0)$

d $(2, -3)$

e Vertical stretching by a factor of 3, translation of 2 units to the right and 3 units down

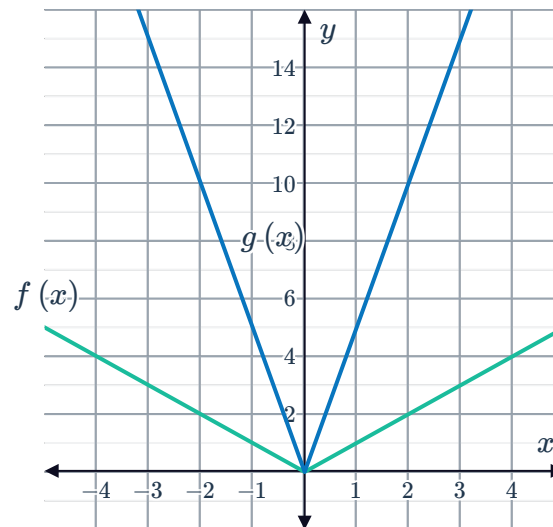
f $f(x) = 3|x - 2| - 3$

31

a

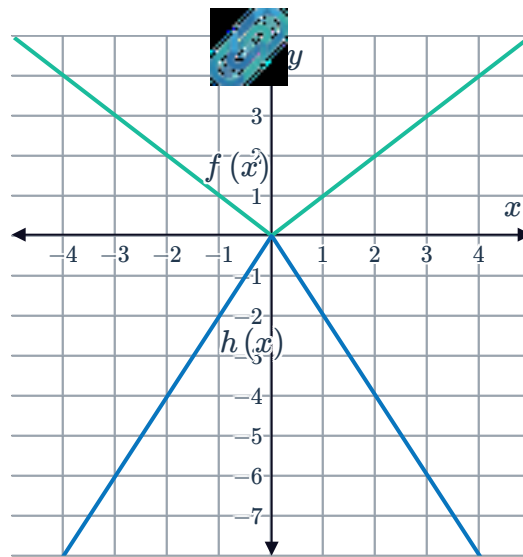
x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3
$g(x) = 5f(x)$	15	10	5	0	5	10	15
$h(x) = -2f(x)$	-6	-4	-2	0	-2	-4	-6

b



c vertically stretched by a factor of 5

d



e Vertically stretched by a factor of 2 and reflected about the x -axis

Applications

32

a 80 mi

b 8 minutes

c Slower

33

a $0 \leq x \leq 60$

b 120 km

c 30 minutes

d Faster, because the lines are steeper than in the first graph.

34

a 700 mi/h

b $t = \left| \frac{3}{35}x - 90 \right|$

c 90 minutes

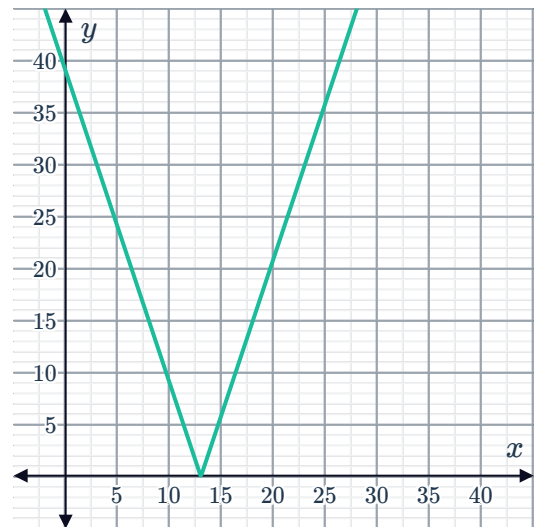
d 2.5 hours

35

a $y = 3|x - 13|$

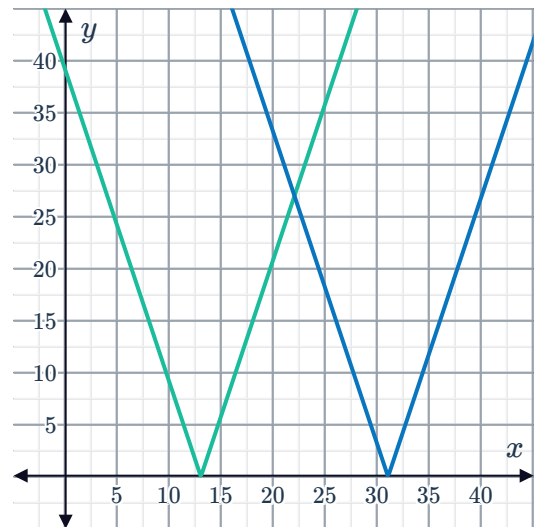


b



c $y = 3|x - 31|$

d



e

- i Yes ii No iii Yes iv No